

ZACH CUTLER

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EDUCATION

University of Utah, GPA 3.77

B.S./M.S., Computer Science

Spring 2021

Minor in Cognitive Science

Dean's List: Fall 2015, Spring 2016, Fall 2016, Spring 2017, Fall 2017, Fall 2018, Spring 2019, Fall 2019, Spring 2020

WORK EXPERIENCE

Software Developer, Datavisyn

Summer 2021 - Present

- Helped design and develop multiple applications used by leading pharmaceutical companies
- Re designed company's flagship product from ground up
- Created custom data visualisation approaches for various drug development problems

Research Assistant, Scientific Computing and Imaging Institute, University of Utah

Fall 2020-Summer 2021

- Worked with Professor Alexander Lex and the Visualisation Design Lab
- Helped design and create multiple tools used for academic research
- Assisted in the writing process for full length papers, short length papers, and blog posts

Teaching Assistant, School of Computing, University of Utah

CS 3505 - Software Practice II

Fall 2018

- Helped students learn C++ and develop good programming habits
- Assisted students in group work by modeling and teaching collaborative work skills
- Taught two weekly lab sessions to help students apply course material

CS 1410 - Intro to Object Oriented Programming

Spring 2018; Spring 2019

- Helped students learn Java and basic object oriented programming
- Evaluated student homework, labs and exams and provided feedback

Lead Instructor, Graphics & Robotics Exploration Camp, University of Utah

Summer 2018

- Taught AP Computer Science Preparation and iOS App Development with Swift
- Led the teaching team for the courses to provide daily instruction
- Coordinated curriculum development and planning for both courses

PROJECTS

Ttrack

vdl.sci.utah.edu/publications/2020_visshort_ttrack

- Helped create a provenance tracking library and a sister library to visualise provenance.
- Presented video and Q&A at IEEE VIS; youtu.be/bi6FfsWV_9k?t=3149
- Presented to the Gehlenborg Lab at Harvard Medical School
- Co-authored blog post to explain our research to a wider audience; vdl.sci.utah.edu/blog/2020/10/28/ttrack/

Intent

vdl.sci.utah.edu/publications/2020_preprint_intent

- Helped create a crowd sourced user study utilising the Ttrack library
- Integrated Ttrack-Vis into a complex system

reVISit

vdl.sci.utah.edu/reVISit/

- Developed a playback feature that allows users to watch how participants in user studies interacted with their visualisation step by step

PUBLICATIONS

Ttrack: A Library for Provenance Tracking in Web-Based Visualizations.

Cutler, Z., Gadhave, K., & Lex, A. Ttrack: A Library for Provenance Tracking in Web-Based Visualizations. IEEE Visualization Conference (VIS), 116-120, doi:10.1109/VIS47514.2020.00030, 2020.

Predicting Intent Behind Selections in Scatterplot Visualizations

Kiran Gadhave, Jochen Görtler, Zach Cutler, Carolina Nobre, Oliver Deussen, Miriah Meyer, Jeff Phillips, Alexander Lex, Predicting Intent Behind Selections in Scatterplot Visualizations. Information Visualization, 20(4): 207-228, doi:10.1177/14738716211038604, 2021.

re-VISit: Looking Under the Hood of Interactive Visualization Studies

Carolina Nobre, Dylan Wootton, Zach Cutler, Lane Harrison, Hanspeter Pfister, Alexander Lex, re-VISit: Looking Under the Hood of Interactive Visualization Studies. SIGCHI Conference on Human Factors in Computing Systems (CHI), 1-13, doi:10.1145/3411764.3445382, 2021.

Reusing Interactive Analysis Workflows.

Kiran Gadhave, Zach Cutler, Alexander Lex, Reusing Interactive Analysis Workflows Computer Graphics Forum (EuroVis), 41(3): 133-144, doi:10.1111/cgf.14528, 2022.